

5-7

1. Suppose that the function f has a continuous second derivative for all x , and that $f(0) = 2$, $f'(0) = -3$, and $f''(0) = 0$. Let g be a function whose derivative is given by $g'(x) = e^{-2x}(3f(x) + 2f'(x))$ for all x .
- (a) Write an equation of the line tangent to the graph of f at the point where $x = 0$.
- (b) Is there sufficient information to determine whether or not the graph of f has a point of inflection when $x = 0$? Explain your answer.
- (c) Given that $g(0) = 4$, write an equation of the line tangent to the graph of g at the point where $x = 0$.
- (d) Show that $g''(x) = e^{-2x}(-6f(x) - f'(x) + 2f''(x))$. Does g have a local maximum at $x = 0$? Justify your answer.

Part A

1 point is earned for the equation

Slope at $x = 0$ is $f'(0) = -3$

At $x = 0$, $y = 2$

$$y - 2 = -3(x - 0)$$



0	1
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The student response earns all of the following points:

1 point is earned for the equation

Slope at $x = 0$ is $f'(0) = -3$

At $x = 0$, $y = 2$

$$y - 2 = -3(x - 0)$$

Part B

1 point is earned for the answer

1 point is earned for the explanation

No, Whether $f''(x)$ changes sign at $x = 0$ is unknown. The only given value of $f''(x)$ is $f''(0) = 0$.



0	1	2
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The student response earns all of the following points:

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1 point is earned for the answer

1 point is earned for the explanation

No, Whether $f''(x)$ changes sign at $x = 0$ is unknown. The only given value of $f''(x)$ is $f''(0) = 0$.**Part C**1 point is earned for: $g'(0)$

1 point is earned for the equation

$$g'(x) = e^{-2x} (3f(x) + 2f'(x))$$

$$\begin{aligned} g'(0) &= e^0 (3f(0) + 2f'(0)) \\ &= 3(2) + 2(-3) = 0 \end{aligned}$$

$$y - 4 = 0(x - 0)$$

$$y = 4$$



0	1	2
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The student response earns all of the following points:

1 point is earned for: $g'(0)$

1 point is earned for the equation

$$g'(x) = e^{-2x} (3f(x) + 2f'(x))$$

$$\begin{aligned} g'(0) &= e^0 (3f(0) + 2f'(0)) \\ &= 3(2) + 2(-3) = 0 \end{aligned}$$

$$y - 4 = 0(x - 0)$$

$$y = 4$$

Part D

2 points are earned for verify derivative 0/2 product or chain rule error

<-1> algebra errors

1 point is earned for: $g'(0) = 0$ and $g''(0)$

1 point is earned for the answer and reasoning

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$$g'(x) = e^{-2x} (3f(x) + 2f'(x))$$

$$g''(x) = (-2e^{-2x})(3f(x) + 2f'(x)) + e^{-2x} (3f'(x) + 2f''(x))$$

$$= e^{-2x} (-6f(x) + f'(x) + 2f''(x))$$

$$g''(0) = e^0 [(-6)(2) - (-3) + 2(0)] = -9$$

Since $g'(0) = 0$ and $g''(0) < 0$, g does have a local maximum at $x = 0$.



0	1	2	3	4
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The student response earns all of the following points:

2 points are earned for verify derivative 0/2 product or chain rule error

<-1> algebra errors

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$$g'(x) = e^{-2x} (3f(x) + 2f'(x))$$

$$g''(x) = (-2e^{-2x})(3f(x) + 2f'(x)) + e^{-2x} (3f'(x) + 2f''(x))$$

$$= e^{-2x} (-6f(x) + f'(x) + 2f''(x))$$

$$g''(0) = e^0 [(-6)(2) - (-3) + 2(0)] = -9$$

Since $g'(0) = 0$ and $g''(0) < 0$, g does have a local maximum at $x = 0$.

5-7

2. Suppose that the function f has a continuous second derivative for all x , and that $f(0) = 2$, $f'(0) = -3$, and $f''(0) = 0$. Let g be a function whose derivative is given by $g'(x) = e^{-2x} (3f(x) + 2f'(x))$ for all x . Show that $g''(x) = e^{-2x} (-6f(x) - f'(x) + 2f''(x))$. Does g have a local maximum at $x = 0$? Justify your answer.

Part D

2 points are earned for verify derivative 0/2 product or chain rule error

< -1 > algebra errors

1 point is earned for: $g'(0) = 0$ and $g''(0)$

1 point is earned for the answer and reasoning

$$g'(x) = e^{-2x} (3f(x) + 2f'(x))$$

$$g''(x) = (-2e^{-2x}) (3f(x) + 2f'(x)) + e^{-2x} (3f'(x) + 2f''(x))$$

$$= e^{-2x} (-6f(x) - f'(x) + 2f''(x))$$

$$g''(0) = e^0 [(-6)(2) - (-3) + 2(0)] = -9$$

Since $g'(0) = 0$ and $g''(0) < 0$, g does have a local maximum at $x = 0$.



0	1	2	3	4
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The student response earns all of the following points:

2 points are earned for verify derivative 0/2 product or chain rule error

< -1 > algebra errors

1 point is earned for: $g'(0) = 0$ and $g''(0)$

1 point is earned for the answer and reasoning

$$g'(x) = e^{-2x} (3f(x) + 2f'(x))$$

$$g''(x) = (-2e^{-2x}) (3f(x) + 2f'(x)) + e^{-2x} (3f'(x) + 2f''(x))$$

$$= e^{-2x} (-6f(x) - f'(x) + 2f''(x))$$

$$g''(0) = e^0 [(-6)(2) - (-3) + 2(0)] = -9$$

Since $g'(0) = 0$ and $g''(0) < 0$, g does have a local maximum at $x = 0$.

5-7

3. A GRAPHING CALCULATOR IS REQUIRED FOR THIS QUESTION.

You are permitted to use your calculator to solve an equation, find the derivative of a function at a point, or calculate the value of a definite integral. However, you must clearly indicate the setup of your question, namely the equation, function, or integral you are using. If you use other built-in features or programs, you must show the mathematical steps necessary to produce your results. Your work must be expressed in standard mathematical notation rather than calculator syntax.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

Let f be a twice-differentiable function such that $f'(2) = 0$. The second derivative of f is given by $f''(x) = x^2e^{2-x} - 1$ for $0 \leq x \leq 6$.

- On what open intervals contained in $0 < x < 6$ is the graph of f concave down? Give a reason for your answer.
- Does f have a relative minimum, a relative maximum, or neither at $x = 2$? Justify your answer.
- Use the Mean Value Theorem on the closed interval $[2, 4]$ to show that $f'(4)$ cannot equal 8.5.
- Does the graph of f have a point of inflection at $x = 5$? Give a reason for your answer.

Part A

A maximum of 1 out of 2 points is earned for one correct interval with reason and no incorrect intervals.

Select a point value to view scoring criteria, solutions, and/or examples to score the response.



0	1	2
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The student response accurately includes both of the criteria below.

- ☐ intervals
- ☐ reason

Solution:

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The graph of f is concave down on $(0, 0.464)$ (or $(0, 0.463)$) and $(5.357, 6)$ (or $(5.356, 6)$) because $f''(x)$ is negative on those intervals.

Part B

Note: Sign charts are a useful tool to investigate and summarize the behavior of a function. By itself a sign chart for $f'(x)$ or $f''(x)$ is not a sufficient response for a justification.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2
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The student response accurately includes both of the criteria below.

- ☐ answer
- ☐ justification

Solution:

Because $f'(2) = 0$ and $f''(2) = 3 > 0$, by the Second Derivative Test f has a relative minimum at $x = 2$.

Part C

The first point requires use of average rate of change of f' on $[2, 4]$.

The second point is earned for using MVT to conclude that there would be a number c in the interval $(2, 4)$ such that $f''(c) = 4.25$. The third point is earned for showing that this is not possible.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2	3
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The student response accurately includes all three of the criteria below.

- ☐ considers average rate of change
- ☐ uses Mean Value Theorem to determine $f''(c) = 4.25$
- ☐ conclusion of “not possible” using Mean Value Theorem

Solution:

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If $f'(4) = 8.5$, then the average rate of change of f' on the closed interval $[2, 4]$ would be

$$\frac{f'(4) - f'(2)}{4 - 2} = \frac{8.5 - 0}{2} = 4.25.$$

Since f' is continuous on $[2, 4]$ and differentiable on $(2, 4)$, by the Mean Value Theorem there would be a number c in the interval $(2, 4)$ such that $f''(c) = 4.25$.

However, since $f''(x) = x^2e^{2-x} - 1$, it follows that $f''(x) = 4.25$ has no solution on the interval $[2, 4]$. Therefore, there is no number c in the interval $(2, 4)$ such that $f''(c) = 4.25$.

Thus, it is not possible that $f'(4) = 8.5$.

Part D

Note: Sign charts are a useful tool to investigate and summarize the behavior of a function. By itself a sign chart for $f'(x)$ or $f''(x)$ is not a sufficient response for a justification.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2
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The student response accurately includes both of the criteria below.

- ☐ answer
- ☐ reason

Solution:

Because $f''(x)$ does not change sign at $x = 5$, the graph of f does not have a point of inflection there. $f''(x) > 0$ on $(0.464, 5.357)$.

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4. Consider the curve given by $x^2 + 4y^2 = 7 + 3xy$.

(a) Show that $\frac{dy}{dx} = \frac{3y-2x}{8y-3x}$.

(b) Show that there is a point P with x -coordinate 3 at which the line tangent to the curve at P is horizontal. Find the y -coordinate of P .

(c) Find the value of $\frac{d^2y}{dx^2}$ at the point P found in part (b). Does the curve have a local maximum, a local minimum, or neither at the point P ? Justify your answer.

Part A

1 point is earned for implicit differentiation

1 point is earned for solving for y'

$$\begin{aligned} 2x + 8yy' &= 3y + 3xy' \\ (8y - 3x)y' &= 3y - 2x \\ y' &= \frac{3y-2x}{8y-3x} \end{aligned}$$



0	1	2
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The student response earns all of the following points:

1 point is earned for implicit differentiation

1 point is earned for solving for y'

$$\begin{aligned} 2x + 8yy' &= 3y + 3xy' \\ (8y - 3x)y' &= 3y - 2x \\ y' &= \frac{3y-2x}{8y-3x} \end{aligned}$$

Part B

1 point is earned for: $\frac{dy}{dx} = 0$

1 point is earned for showing slope is 0 at (3, 2)

1 point is earned for showing (3, 2) lies on curve

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$$\frac{3y - 2x}{8y - 3x} = 0; 3y - 2x = 0$$

When $x = 3$, $3y = 6$

$$y = 2$$

$$3^2 + 4 \cdot 2^2 = 25 \text{ and } 7 + 3 \cdot 3 \cdot 2 = 25$$

Therefore, $P = (3, 2)$ is on the curve and the slope is 0 at this point.



0	1	2	3
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The student response earns all of the following points:

1 point is earned for: $\frac{dy}{dx} = 0$

1 point is earned for showing slope is 0 at $(3, 2)$

1 point is earned for showing $(3, 2)$ lies on curve

$$\frac{3y - 2x}{8y - 3x} = 0; 3y - 2x = 0$$

When $x = 3$, $3y = 6$

$$y = 2$$

$$3^2 + 4 \cdot 2^2 = 25 \text{ and } 7 + 3 \cdot 3 \cdot 2 = 25$$

Therefore, $P = (3, 2)$ is on the curve and the slope is 0 at this point.

Part C

2 points are earned for: $\frac{d^2y}{dx^2}$

1 point is earned for the value of $\frac{d^2y}{dx^2}$ at $(3, 2)$

1 point is earned for the conclusion with justification

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$$\frac{d^2y}{dx^2} = \frac{(8y-3x)(3y'-2)-(3y-2x)(8y'-3)}{(8y-3x)^2}$$

$$\text{At } P = (3, 2), \frac{d^2y}{dx^2} = \frac{(16-9)(-2)}{(16-9)^2} = -\frac{2}{7}.$$

Since $y' = 0$ and $y'' < 0$ at P , the curve has a local maximum at P .



0	1	2	3	4
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The student response earns all of the following points:

2 points are earned for: $\frac{d^2y}{dx^2}$

1 point is earned for the value of $\frac{d^2y}{dx^2}$ at $(3, 2)$

1 point is earned for the conclusion with justification

$$\frac{d^2y}{dx^2} = \frac{(8y-3x)(3y'-2)-(3y-2x)(8y'-3)}{(8y-3x)^2}$$

$$\text{At } P = (3, 2), \frac{d^2y}{dx^2} = \frac{(16-9)(-2)}{(16-9)^2} = -\frac{2}{7}.$$

Since $y' = 0$ and $y'' < 0$ at P , the curve has a local maximum at P .

5-7

5. Consider the curve given by $x^2 + 4y^2 = 7 + 3xy$.

Find the value of $\frac{d^2y}{dx^2}$ at the point P found in part (b). Does the curve have a local maximum, a local minimum, or neither at the point P ? Justify your answer.

Part C

2 points are earned for: $\frac{d^2y}{dx^2}$

1 point is earned for the value of $\frac{d^2y}{dx^2}$ at $(3, 2)$

1 point is earned for the conclusion with justification

$$\frac{d^2y}{dx^2} = \frac{(8y-3x)(3y'-2) - (3y-2x)(8y'-3)}{(8y-3x)^2}$$

$$\text{At } P = (3, 2), \frac{d^2y}{dx^2} = \frac{(16-9)(-2)}{(16-9)^2} = -\frac{2}{7}.$$

Since $y' = 0$ and $y'' < 0$ at P , the curve has a local maximum at P .



0	1	2	3	4
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The student response earns all of the following points:

2 points are earned for: $\frac{d^2y}{dx^2}$

1 point is earned for the value of $\frac{d^2y}{dx^2}$ at $(3, 2)$

1 point is earned for the conclusion with justification

$$\frac{d^2y}{dx^2} = \frac{(8y-3x)(3y'-2) - (3y-2x)(8y'-3)}{(8y-3x)^2}$$

$$\text{At } P = (3, 2), \frac{d^2y}{dx^2} = \frac{(16-9)(-2)}{(16-9)^2} = -\frac{2}{7}.$$

Since $y' = 0$ and $y'' < 0$ at P , the curve has a local maximum at P .

5-7

6. Consider the differential equation $\frac{dy}{dx} = \frac{3-x}{y}$.

Let $y = f(x)$ be the particular solution to the given differential equation for $1 < x < 5$ such that the line $y = -2$ is tangent to the graph of f . Find the x -coordinate of the point of tangency, and determine whether f has a local maximum, local minimum, or neither at this point. Justify your answer.

Part A

1 point is earned for: $x = 3$

1 point is earned for the local minimum

1 point is earned for the justification

$$\frac{dy}{dx} = 0 \text{ when } x = 3$$

$$\left. \frac{d^2y}{dx^2} \right|_{(3,-2)} = \left. \frac{-y-y'(3-x)}{y^2} \right|_{(3,-2)} = \frac{1}{2},$$

so f has a local minimum at this point.

or

Because f is continuous for $1 < x < 5$, there is an interval containing $x = 3$ on which $y < 0$. On this interval, $\frac{dy}{dx}$ is negative to the left of $x = 3$ and $\frac{dy}{dx}$ is positive to the right of $x = 3$. Therefore f has a local minimum at $x = 3$.



0	1	2	3
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The student response earns all of the following points:

1 point is earned for: $x = 3$

1 point is earned for the local minimum

1 point is earned for the justification

$$\frac{dy}{dx} = 0 \text{ when } x = 3$$

$$\left. \frac{d^2y}{dx^2} \right|_{(3,-2)} = \left. \frac{-y-y'(3-x)}{y^2} \right|_{(3,-2)} = \frac{1}{2},$$

so f has a local minimum at this point.

or

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Because f is continuous for $1 < x < 5$, there is an interval containing $x = 3$ on which $y < 0$. On this interval, $\frac{dy}{dx}$ is negative to the left of $x = 3$ and $\frac{dy}{dx}$ is positive to the right of $x = 3$. Therefore f has a local minimum at $x = 3$.

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7. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Consider the differential equation $\frac{dy}{dx} = -1 + \frac{y^2}{x}$.

(a) Show that $\frac{d^2y}{dx^2} = \frac{2y^3 - y^2 - 2xy}{x^2}$.

(b) Let $y = g(x)$ be the particular solution to the differential equation $\frac{dy}{dx} = -1 + \frac{y^2}{x}$ with initial condition $g(4) = 2$. Does g have a relative minimum, a relative maximum, or neither at $x = 4$? Justify your answer.

(c) Let $y = h(x)$ be the particular solution to the differential equation $\frac{dy}{dx} = -1 + \frac{y^2}{x}$ with initial condition $h(1) = 2$. Write the second-degree Taylor polynomial for h about $x = 1$.

(d) For the function h given in part (c), it is known that $|h'''(x)| \leq 60$ for all x in the interval $0.9 \leq x \leq 1.1$. Let A represent the approximation of $h(1.1)$ found by using the second-degree Taylor polynomial for h about $x = 1$ from part (c). Use the Lagrange error bound to show that A differs from $h(1.1)$ by at most 0.01.

Part A**1 point is earned for:** quotient rule**1 point is earned for:** $\frac{d^2y}{dx^2}$

$$\begin{aligned}\frac{d^2y}{dx^2} &= \frac{x \cdot 2y \frac{dy}{dx} - y^2 \cdot 1}{x^2} \\ &= \frac{2xy \left(-1 + \frac{y^2}{x}\right) - y^2}{x^2} = \frac{2y^3 - y^2 - 2xy}{x^2}\end{aligned}$$



0	1	2
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The student response earns all of the following points:

1 point is earned for: quotient rule**1 point is earned for:** $\frac{d^2y}{dx^2}$

$$\begin{aligned}\frac{d^2y}{dx^2} &= \frac{x \cdot 2y \frac{dy}{dx} - y^2 \cdot 1}{x^2} \\ &= \frac{2xy \left(-1 + \frac{y^2}{x}\right) - y^2}{x^2} = \frac{2y^3 - y^2 - 2xy}{x^2}\end{aligned}$$

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Part B

1 point is earned for: considers $\frac{dy}{dx}\big|_{(x,y)=(4,2)}$

1 point is earned for: answer with justification

$$\frac{dy}{dx}\big|_{(x,y)=(4,2)} = -1 + \frac{4}{4} = 0$$

$$\frac{d^2y}{dx^2}\big|_{(x,y)=(4,2)} = \frac{2 \cdot 8 - 4 - 16}{16} = -\frac{1}{4} < 0$$

By the Second Derivative Test, g has a relative maximum at $x = 4$.



0	1	2
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The student response earns all of the following points:

1 point is earned for: considers $\frac{dy}{dx}\big|_{(x,y)=(4,2)}$

1 point is earned for: answer with justification

$$\frac{dy}{dx}\big|_{(x,y)=(4,2)} = -1 + \frac{4}{4} = 0$$

$$\frac{d^2y}{dx^2}\big|_{(x,y)=(4,2)} = \frac{2 \cdot 8 - 4 - 16}{16} = -\frac{1}{4} < 0$$

By the Second Derivative Test, g has a relative maximum at $x = 4$.

Part C

1 point is earned for: uses $\frac{dy}{dx}\big|_{(x,y)=(1,2)}$ and $\frac{d^2y}{dx^2}\big|_{(x,y)=(1,2)}$

2 points are earned for: Taylor polynomial

$$\frac{dy}{dx}\big|_{(x,y)=(1,2)} = -1 + \frac{4}{1} = 3$$

$$\frac{d^2y}{dx^2}\big|_{(x,y)=(1,2)} = \frac{2 \cdot 8 - 4 - 4}{1} = 8$$

The second-degree Taylor polynomial for h about $x = 1$ is

$$T_2(x) = 2 + 3(x - 1) + \frac{8}{2!}(x - 1)^2 = 2 + 3(x - 1) + 4(x - 1)^2.$$

5-7



0	1	2	3
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The student response earns all of the following points:

1 point is earned for: uses $\left. \frac{dy}{dx} \right|_{(x,y)=(1,2)}$ and $\left. \frac{d^2y}{dx^2} \right|_{(x,y)=(1,2)}$

2 points are earned for: Taylor polynomial

$$\left. \frac{dy}{dx} \right|_{(x,y)=(1,2)} = -1 + \frac{4}{1} = 3$$

$$\left. \frac{d^2y}{dx^2} \right|_{(x,y)=(1,2)} = \frac{2 \cdot 8 - 4 - 4}{1} = 8$$

The second-degree Taylor polynomial for h about $x = 1$ is

$$T_2(x) = 2 + 3(x - 1) + \frac{8}{2!}(x - 1)^2 = 2 + 3(x - 1) + 4(x - 1)^2.$$

Part D

1 point is earned for: form of the error bound

1 point is earned for: analysis

$$|h(1.1) - A| \leq \frac{\max_{1.0 \leq x \leq 1.1} |h'''(x)| |1.1 - 1|^3}{3!} \leq \frac{60}{6} \cdot \frac{1}{1000} = \frac{10}{1000} = \frac{1}{100}$$



0	1	2
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The student response earns all of the following points:

1 point is earned for: form of the error bound

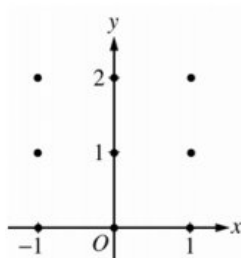
1 point is earned for: analysis

$$|h(1.1) - A| \leq \frac{\max_{1.0 \leq x \leq 1.1} |h'''(x)| |1.1 - 1|^3}{3!} \leq \frac{60}{6} \cdot \frac{1}{1000} = \frac{10}{1000} = \frac{1}{100}$$

5-7

8. Consider the differential equation $\frac{dy}{dx} = \frac{1}{2}x + y - 1$.

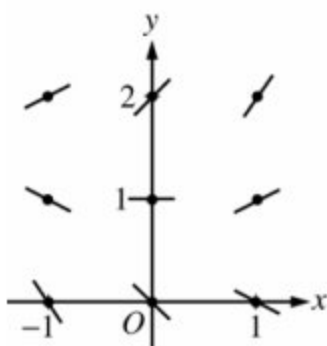
(a) On the axes provided, sketch a slope field for the given differential equation at the nine points indicated.



- (b) Find $\frac{d^2y}{dx^2}$ in terms of x and y . Describe the region in the xy -plane in which all solution curves to the differential equation are concave up.
- (c) Let $y = f(x)$ be a particular solution to the differential equation with the initial condition $f(0) = 1$. Does f have a relative minimum, a relative maximum, or neither at $x = 0$? Justify your answer.
- (d) Find the values of the constants m and b , for which $y = mx + b$ is a solution to the differential equation.

Part A

2 points are earned for sign of each slope at each point and relative steepness of slope lines in rows and columns



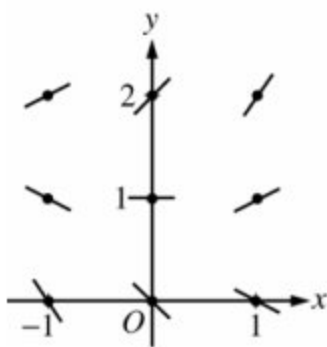
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0	1	2
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The student response earns all of the following points:

2 points are earned for sign of each slope at each point and relative steepness of slope lines in rows and columns

5-7

**Part B**

2 points are earned for $\frac{d^2y}{dx^2}$

1 point is earned for the description

$$\frac{d^2y}{dx^2} = \frac{1}{2} + \frac{dy}{dx} = \frac{1}{2}x + y - \frac{1}{2}$$

Solution curves will be concave up on the half-plane above the line

$$y = -\frac{1}{2}x + \frac{1}{2}.$$



0	1	2	3
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The student response earns all of the following points:

2 points are earned for $\frac{d^2y}{dx^2}$

1 point is earned for the description

$$\frac{d^2y}{dx^2} = \frac{1}{2} + \frac{dy}{dx} = \frac{1}{2}x + y - \frac{1}{2}$$

Solution curves will be concave up on the half-plane above the line

$$y = -\frac{1}{2}x + \frac{1}{2}.$$

Part C

1 point is earned for the answer

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1 point is earned for the justification

$$\frac{dy}{dx}\bigg|_{(0,1)} = 0 + 1 - 1 = 0 \text{ and } \frac{d^2y}{dx^2}\bigg|_{(0,1)} = 0 + 1 - \frac{1}{2} > 0$$

Thus, f has a relative minimum at $(0,1)$.



0	1	2
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The student response earns all of the following points:

1 point is earned for the answer

1 point is earned for the justification

$$\frac{dy}{dx}\bigg|_{(0,1)} = 0 + 1 - 1 = 0 \text{ and } \frac{d^2y}{dx^2}\bigg|_{(0,1)} = 0 + 1 - \frac{1}{2} > 0$$

Thus, f has a relative minimum at $(0,1)$.

Part D

1 point is earned for the value of m

1 point is earned for the value of b

Substituting $y = mx + b$ into the differential equation:

$$m = \frac{1}{2}x + (mx + b) - 1 = (m + \frac{1}{2})x + (b - 1)$$

$$\text{Then } 0 = m + \frac{1}{2} \text{ and } m = b - 1: m = -\frac{1}{2} \text{ and } b = \frac{1}{2}.$$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the value of m

1 point is earned for the value of b

Substituting $y = mx + b$ into the differential equation:

5-7

$$m = \frac{1}{2}x + (mx + b) - 1 = (m + \frac{1}{2})x + (b - 1)$$

Then $0 = m + \frac{1}{2}$ and $m = b - 1$: $m = -\frac{1}{2}$ and $b = \frac{1}{2}$.

9. Consider the differential equation $\frac{dy}{dx} = \frac{1}{2}x + y - 1$

Let $y=f(x)$ be a particular solution to the differential equation with the initial condition $f(0)=1$. Does f have a relative minimum, a relative maximum, or neither at $x=0$? Justify your answer.

Part C

1 point is earned for the answer

1 point is earned for the justification

$$\left. \frac{dy}{dx} \right|_{(0,1)} = 0 + 1 - 1 = 0 \text{ and } \left. \frac{d^2y}{dx^2} \right|_{(0,1)} = 0 + 1 - \frac{1}{2} > 0 \text{ Thus, } f \text{ has a relative minimum at } (0,1) .$$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the answer

1 point is earned for the justification

$$\left. \frac{dy}{dx} \right|_{(0,1)} = 0 + 1 - 1 = 0 \text{ and } \left. \frac{d^2y}{dx^2} \right|_{(0,1)} = 0 + 1 - \frac{1}{2} > 0 \text{ Thus, } f \text{ has a relative minimum at } (0,1) .$$

5-7

10. Consider the differential equation $\frac{dy}{dx} = \frac{3-x}{y}$.

(a) Let $y = f(x)$ be the particular solution to the given differential equation for $1 < x < 5$ such that the line $y = -2$ is tangent to the graph of f . Find the x -coordinate of the point of tangency, and determine whether f has a local maximum, local minimum, or neither at this point. Justify your answer.

(b) Let $y = g(x)$ be the particular solution to the given differential equation for $-2 < x < 8$, with the initial condition $g(6) = -4$. Find $y = g(x)$.

Part A

1 point is earned for: $x = 3$

1 point is earned for the local minimum

1 point is earned for the justification

$$\frac{dy}{dx} = 0 \text{ when } x = 3$$

$$\left. \frac{d^2y}{dx^2} \right|_{(3,-2)} = \left. \frac{-y-y'(3-x)}{y^2} \right|_{(3,-2)} = \frac{1}{2},$$

so f has a local minimum at this point.

or

Because f is continuous for $1 < x < 5$, there is an interval containing $x = 3$ on which $y < 0$. On this interval, $\frac{dy}{dx}$ is negative to the left of $x = 3$ and $\frac{dy}{dx}$ is positive to the right of $x = 3$. Therefore f has a local minimum at $x = 3$.



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for: $x = 3$

1 point is earned for the local minimum

1 point is earned for the justification

$$\frac{dy}{dx} = 0 \text{ when } x = 3$$

$$\left. \frac{d^2y}{dx^2} \right|_{(3,-2)} = \left. \frac{-y-y'(3-x)}{y^2} \right|_{(3,-2)} = \frac{1}{2},$$

5-7

so f has a local minimum at this point.

or

Because f is continuous for $1 < x < 5$, there is an interval containing $x = 3$ on which $y < 0$. On this interval, $\frac{dy}{dx}$ is negative to the left of $x = 3$ and $\frac{dy}{dx}$ is positive to the right of $x = 3$. Therefore f has a local minimum at $x = 3$.

Part B

1 point is earned for the separates variables

1 point is earned for the antiderivative of dy term

1 point is earned for the antiderivative of dx term

1 point is earned for the constant of integration

1 point is earned for the uses initial condition $g(6) = -4$

1 point is earned for solving for y

$$y \, dy = (3 - x) \, dx$$

$$\frac{1}{2}y^2 = 3x - \frac{1}{2}x^2 + C$$

$$8 = 18 - 18 + C; C = 8$$

$$y^2 = 6x - x^2 + 16$$

$$y = -\sqrt{6x - x^2 + 16}$$



0	1	2	3	4	5	6
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The student response earns all of the following points:

1 point is earned for the separates variables

1 point is earned for the antiderivative of dy term

1 point is earned for the antiderivative of dx term

1 point is earned for the constant of integration

1 point is earned for the uses initial condition $g(6) = -4$

1 point is earned for solving for y

5-7

$$y \, dy = (3 - x)dx$$

$$\frac{1}{2}y^2 = 3x - \frac{1}{2}x^2 + C$$

$$8 = 18 - 18 + C; C = 8$$

$$y^2 = 6x - x^2 + 16$$

$$y = -\sqrt{6x - x^2 + 16}$$

5-7

11. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

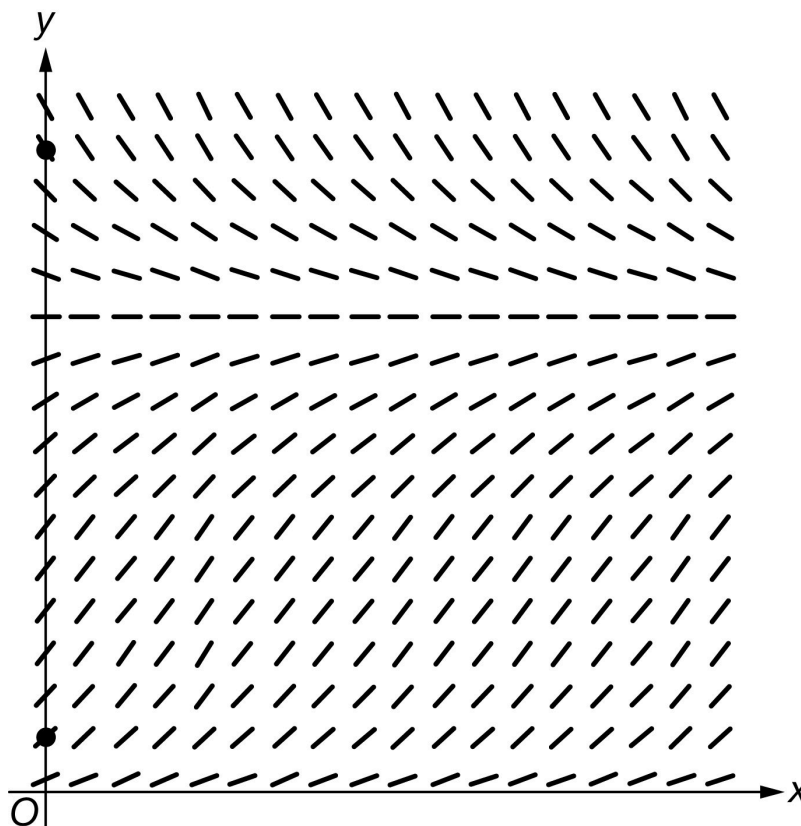
Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

Consider the differential equation $\frac{dy}{dx} = \frac{-y}{4} \ln\left(\frac{y}{50}\right)$.

(a) A portion of the slope field for the differential equation is given below. Sketch the solution curves through the two indicated points for $x \geq 0$.



(b) Find $\lim_{y \rightarrow 50} \frac{dy}{dx}$.

(c) Let f be the particular solution to the differential equation with initial condition $f(0) = 20$. Find $f'(0)$, and explain why f is increasing for all x .

(d) For $0 < y < 40$, at what value of y does $\frac{dy}{dx}$ attain its maximum value? Justify your answer.

5-7

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Each solution curve must pass through the appropriate point, extend reasonably close to the right edge of the square, and have no obvious conflicts with the given slope lines.

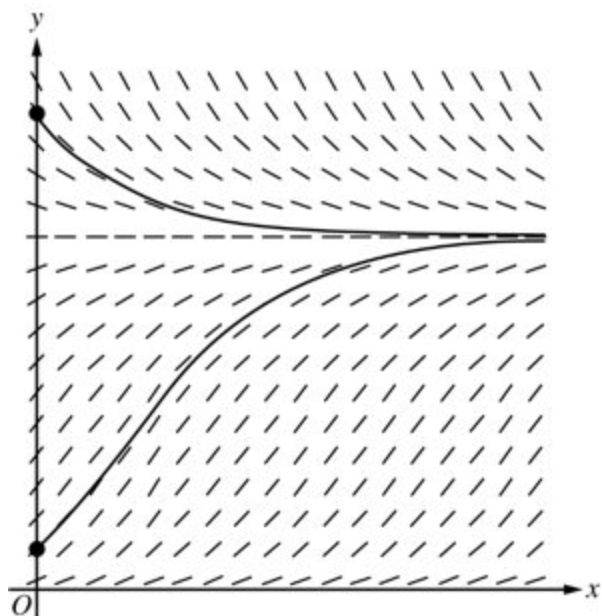
Ignore any portion of a solution curve that extends beyond the square.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Top solution curve
- ☐ Bottom solution curve

Solution:**Part B**

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

An answer of 0 earns the point.

The answer of $-\frac{50}{4} \ln\left(\frac{50}{50}\right)$ need not be simplified, but any subsequent simplification must be correct.

5-7

Limit notation is not required.

Expressions that equate the limit with the derivative such as $\lim_{y \rightarrow 50} \frac{dy}{dx} = \frac{-y}{4} \ln\left(\frac{y}{50}\right) = \frac{-50}{4} \ln\left(\frac{50}{50}\right)$, or other linkage errors, will not earn the point, even in the presence of the correct evaluation.

Expressions with incomplete limit notation, such as $\lim_{y \rightarrow 50} = \frac{-50}{4} \ln\left(\frac{50}{50}\right)$, are eligible for the point.

The expression $\lim_{y \rightarrow 50} \frac{-50}{4} \ln\left(\frac{50}{50}\right)$ is not incorrect, but the limit must be resolved to earn this point.



0	1
---	---

The student response accurately includes the criteria below.

☐ Limit

Solution:

$$\lim_{y \rightarrow 50} \frac{dy}{dx} = \lim_{y \rightarrow 50} \frac{-y}{4} \ln\left(\frac{y}{50}\right) = \frac{-50}{4} \ln\left(\frac{50}{50}\right) = 0$$

Part C

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point is earned for $-5 \ln\left(\frac{2}{5}\right)$ or the equivalent; no supporting work is necessary.

The evaluation does not need to be simplified, and incorrect attempts at simplifying a correct expression will not be read.

For the second point to be earned, the response must:

- Argue that $\frac{dy}{dx}$ is positive for $0 < y < 50$, thus f is increasing for $0 < y < 50$.
- Argue that the particular solution cannot cross the line $y = 50$. (This may be done implicitly by restricting the domain of consideration to $0 < y < 50$.)

Explanations may be graphical but must address both of the above points.

Explanations that appeal only to the sign of $f'(0)$ will not earn this point.

5-7



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ $f'(0)$
- ☐ Explanation

Solution:

$$f'(0) = \left. \frac{dy}{dx} \right|_{(x,y)=(0,20)} = \frac{-20}{4} \ln\left(\frac{20}{50}\right) = -5 \ln\left(\frac{2}{5}\right)$$

Because $\frac{dy}{dx} = \frac{-y}{4} \ln\left(\frac{y}{50}\right) > 0$ for $0 < y < 50$, f is increasing provided $f(x) < 50$.

Note that $y = 50$ is also a solution to the differential equation, so the graph of the particular solution $y = f(x)$ with $f(0) = 20$ is below the line $y = 50$ and cannot cross the line $y = 50$.

Thus, f is increasing for all x .

Part D

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point is earned by correctly applying the product rule in finding $\frac{d^2y}{dx^2}$, with or without using the chain rule. For example, $\frac{d^2y}{dx^2} = -\frac{1}{4} \cdot \ln\left(\frac{y}{50}\right) - \frac{y}{4} \cdot \frac{1}{y}$ would earn the first point.

The second point is earned for correctly applying the chain rule (twice) in finding $\frac{d^2y}{dx^2}$. This point can be earned even if an error was made in applying the product rule that resulted in the first point not being earned. In general, expressions of the form $\frac{d^2y}{dx^2} = c \cdot \ln\left(\frac{y}{50}\right) \cdot \frac{dy}{dx} + g(y) \cdot \frac{dy}{dx}$, where $g(y)$ may be a nonzero constant and c is a nonzero constant, earn this point.

For the first two points, if both the factors 50 and $\frac{1}{50}$ are not present in the response, assume that these factors were canceled. Furthermore, in the presence of the factor 50, assume that a missing factor $\frac{1}{50}$ is an error in the application of the chain rule. Thus, $\frac{d^2y}{dx^2} = -\frac{1}{4} \cdot \frac{dy}{dx} \cdot \ln\left(\frac{y}{50}\right) - \frac{y}{4} \cdot \frac{50}{y} \cdot \frac{dy}{dx}$ would earn the first point but not the second point.

For the first two points, each point is earned with the correct expression and any subsequent errors in simplification come off of the fourth point.

5-7

The third point is earned when the declared $\frac{d^2y}{dx^2}$ is set equal to zero and one initial (correct or otherwise) algebraic step towards solving this equation for y is made. However, the declared expression for $\frac{d^2y}{dx^2}$ must contain the variable y and must be distinct from $\frac{dy}{dx} = \frac{-y}{4} \ln\left(\frac{y}{50}\right)$. (This means a score of 0-0-1-0 is possible in this part.)

The fourth point cannot be earned without having earned the first three points.

The fourth point requires a discussion of the behavior of $\frac{d^2y}{dx^2}$ both to the left and right of $y = 50e^{-1}$.

Sign charts, labeled or otherwise, are not read for the fourth point.



0	1	2	3	4
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The student response accurately includes **all four** of the criteria below.

- ☐ Product rule
- ☐ Chain rule
- ☐ Attempts to solve $\frac{d^2y}{dx^2} = 0$ for y
- ☐ Answer with justification

Solution:

$$\begin{aligned}\frac{d^2y}{dx^2} &= -\frac{1}{4} \cdot \frac{dy}{dx} \cdot \ln\left(\frac{y}{50}\right) - \frac{y}{4} \cdot \frac{1}{y} \frac{dy}{dx} \\ &= -\frac{1}{4} \cdot \frac{dy}{dx} \cdot \ln\left(\frac{y}{50}\right) - \frac{y}{4} \cdot \frac{1}{y} \frac{dy}{dx} = -\frac{1}{4} \cdot \frac{dy}{dx} \cdot \left(\ln\left(\frac{y}{50}\right) + 1\right)\end{aligned}$$

$$\frac{d^2y}{dx^2} = 0 \Rightarrow \left(\ln\left(\frac{y}{50}\right) + 1\right) = 0 \Rightarrow y = 50e^{-1}$$

Because $\frac{d^2y}{dx^2} > 0$ for $0 < y < 50e^{-1}$ and $\frac{d^2y}{dx^2} < 0$ for $50e^{-1} < y < 40$, $\frac{dy}{dx}$ attains its maximum value at $y = 50e^{-1}$.

5-7

12. Let f be the function defined by $f(x) = k\sqrt{x} - \ln x$, for $x > 0$, where k is a positive constant. For what value of the constant k does f have a critical point at $x = 1$? For this value of k , determine whether f has a relative minimum, relative maximum, or neither at $x = 1$. Justify your answer.

Part B

1 point is earned for: sets $f'(1) = 0$ or $f'(x) = 0$

1 point is earned for solves for k

1 point is earned for the answer

1 point is earned for the justification

$$f'(1) = \frac{1}{2}k - 1 = 0 \Rightarrow k = 2$$

$$\text{When, } k = 2, f'(1) = 0, \text{ and, } f''(1) = -\frac{1}{2} + 1 > 0.$$

f has a relative minimum value at $x = 1$ by the Second Derivative Test.



0	1	2	3	4
---	---	---	---	---

The student response earns all of the following points:

1 point is earned for: sets $f'(1) = 0$ or $f'(x) = 0$

1 point is earned for solves for k

1 point is earned for the answer

1 point is earned for the justification

$$f'(1) = \frac{1}{2}k - 1 = 0 \Rightarrow k = 2$$

$$\text{When, } k = 2, f'(1) = 0, \text{ and, } f''(1) = -\frac{1}{2} + 1 > 0.$$

f has a relative minimum value at $x = 1$ by the Second Derivative Test.

5-7

13. Let f be a function defined and continuous on the closed interval $[a,b]$. If f has a relative maximum at c and $a < c < b$, which of the following statements must be true?

I. $f'(c)$ exists.

II. If $f'(c)$ exists, then $f'(c) = 0$

III. If $f''(c)$ exists, then $f''(c) \leq 0$.

(A) II only

(B) III only

(C) I and II only

(D) I and III only

(E) II and III only



14. Let f be a twice-differentiable function for all real numbers x . Which of the following additional properties guarantees that f has a relative minimum at $x = c$?

(A) $f'(c) = 0$

(B) $f'(c) = 0$ and $f''(c) < 0$

(C) $f'(c) = 0$ and $f''(c) > 0$



(D) $f'(x) > 0$ for $x < c$ and $f'(x) < 0$ for $x > c$

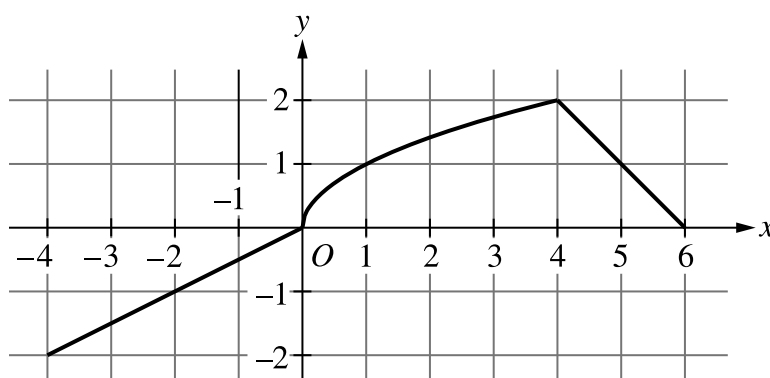
5-7

15. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

Graph of g

The graph of the function g is shown on the interval $[-4, 6]$. The graph of g consists of two line segments and the portion of the graph of $y = \sqrt{x}$ for $0 \leq x \leq 4$. Let f be the function defined by

$$f(x) = g(x) + \int_{-4}^x g(t) \, dt.$$

(a) Find the value of $f(4)$.

(b) Show that f has a critical point at $x = -1$. Use the Second Derivative Test to determine whether f has a local minimum or a local maximum at $x = -1$. Show the work that leads to your answer.

(c) Find the value of $\lim_{x \rightarrow 4^-} \frac{g(x) - g(4)}{x - 4}$.

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

A response does not need to present $\int_{-4}^0 g(t) \, dt$ to earn the first point. The value of ± 4 may be determined directly from the graph.

The inclusion of $g(4)$ is considered in the third point.

5-7

The second point is earned by finding the antiderivative $\frac{2}{3}t^{3/2}$ or by determining the area under this portion of the parabola is $\frac{2}{3} \cdot 8$.

A response that finds an antiderivative of the form $k \cdot t^{3/2}$ where $k \neq 0$ and $k \neq \frac{2}{3}$ does not earn the second point but does earn the third point for the consistent answer of $-2 + k \cdot 8$.

The final answer does not need to be simplified. Any error in simplification does not earn the third point.



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ $\pm$ value of $\int_{-4}^0 g(t) dt$
- ☐ Antiderivative of $\sqrt{t}$
- ☐ Answer

Solution:

$$\begin{aligned}
 f(4) &= g(4) + \int_{-4}^4 g(t) dt = 2 + \int_{-4}^0 g(t) dt + \int_0^4 g(t) dt \\
 &= 2 + (-4) + \int_0^4 \sqrt{t} dt = -2 + \left[\frac{2}{3} t^{3/2} \right]_{t=0}^{t=4} \\
 &= -2 + \frac{2}{3} \cdot 8 = \frac{10}{3}
 \end{aligned}$$

Part B

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point is only earned for the statement $f'(x) = g'(x) + g(x)$ or If there is any equating of unequal quantities, or miscommunication, for example, the response does not earn the first point.

If the first point is not earned, the second point can only be earned for a response that presents

If the first point is earned, a response can earn the second point by stating that

To earn the third point, a response must correctly find or and go on to state that

The fourth point can be earned without any of the first 3 points, provided the response declares a nonzero value or sign of and makes a consistent Second Derivative Test argument based on that value or sign.

5-7



0	1	2	3	4
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The student response accurately includes **all four** of the criteria below.

- ☐ $f'(x) = g'(x) + g(x)$ OR $f'(-1) = g'(-1) + g(-1)$
- ☐ Shows $f'(-1) = 0$
- ☐ $f''(-1)$
- ☐ Answer with reason

Solution:

$$f'(x) = g'(x) + g(x)$$

$$f'(-1) = g'(-1) + g(-1) = \frac{1}{2} + \left(-\frac{1}{2}\right) = 0$$

Because $f'(-1) = 0$, f has a critical point at $x = -1$.

$$f''(x) = g''(x) + g'(x)$$

$$f''(-1) = g''(-1) + g'(-1) = 0 + \frac{1}{2}$$

Because $f'(-1) = 0$ and $f''(-1) > 0$, the Second Derivative Test indicates that f has a local minimum at $x = -1$.

Part C

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The response does not need to state that $\sqrt{x}$ is differentiable.

A minimal response to earn both points is $\frac{1}{2\sqrt{4}}$.

Special case (using L'Hospital's Rule): For $0 \leq x \leq 4$, $g(x) = \sqrt{x}$.

$$\lim_{x \rightarrow 4^-} (\sqrt{x} - g(4)) = 2 - 2 = 0 \text{ and } \lim_{x \rightarrow 4^-} (x - 4) = 0.$$

Because $\sqrt{x}$ is continuous on the interval $[0, 4]$ and differentiable on the interval $(0, 4)$,

$\lim_{x \rightarrow 4^-} [g(x) - g(4)] = 2 - 2 = 0$, and $\lim_{x \rightarrow 4^-} [x - 4] = 4 - 4 = 0$, L'Hospital's Rule may be applied:

5-7

$$\lim_{x \rightarrow 4^-} \frac{g(x) - g(4)}{x - 4} = \lim_{x \rightarrow 4^-} \frac{g'(x)}{1} = \lim_{x \rightarrow 4^-} \frac{\frac{d}{dx} \sqrt{x}}{1} = \lim_{x \rightarrow 4^-} \frac{\frac{1}{2\sqrt{x}}}{1} = \frac{1}{2\sqrt{4}} = \frac{1}{4}.$$

- A response earns the first point by evaluating the limits of the numerator and denominator.
- A response that equates the limit to $\frac{0}{0}$ does not earn the first point but is eligible for the second point. A response may say that the limit is of indeterminate form $\frac{0}{0}$.
- The second point is earned by correctly applying L'Hospital's Rule (by presenting a limit of a ratio of derivatives) and finding the correct answer.



0	1	2
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The student response accurately includes **both** of the criteria below.

- ☐ Uses $g(x) = \sqrt{x}$
- ☐ Answer

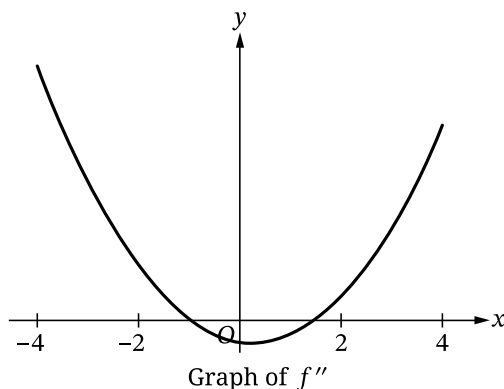
Solution:

For $0 \leq x \leq 4$, $g(x) = \sqrt{x}$.

Because $\sqrt{x}$ has a left-hand derivative at $x = 4$, $\lim_{x \rightarrow 4^-} \frac{g(x) - g(4)}{x - 4} = \frac{d}{dx} \sqrt{x} \Big|_{x=4} = \frac{1}{2\sqrt{x}} \Big|_{x=4} = \frac{1}{4}$.

5-7

16. The function f is twice differentiable, and it is known that $f'(-2) = f'(2) = 0$. The graph of $y = f''(x)$ is shown in the figure.



Which of the following statements about the function f must be true?

- (A) f has a relative maximum at $x = -2$ and a relative minimum at $x = 2$.
(B) f has a relative minimum at $x = -2$ and a relative maximum at $x = 2$.
(C) f has relative minima at $x = -2$ and $x = 2$. ✓
(D) f does not have a relative minimum or a relative maximum at either $x = -2$ or $x = 2$.

Answer C

Correct. Since $f'(-2) = 0$, the function f has a critical point at $x = -2$. This critical point is a relative minimum by the Second Derivative Test because $f''(-2) > 0$. Since $f'(2) = 0$, the function f also has a critical point at $x = 2$. This critical point is also a relative minimum by the Second Derivative Test because $f''(2) > 0$.

5-7

17. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

Consider the function $y = f(x)$ whose curve is given by the equation $2y^2 - 6 = y \sin x$ for $y > 0$.

(a) Show that $\frac{dy}{dx} = \frac{y \cos x}{4y - \sin x}$.

(b) Write an equation for the line tangent to the curve at the point $(0, \sqrt{3})$.

(c) For $0 \leq x \leq \pi$ and $y > 0$, find the coordinates of the point where the line tangent to the curve is horizontal.

(d) Determine whether f has a relative minimum, a relative maximum, or neither at the point found in part (c). Justify your answer.

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point is earned only for correctly implicitly differentiating $2y^2 - 6 = y \sin x$. Responses may use alternative notations for $\frac{dy}{dx}$, such as y' .

The second point may not be earned without the first point.

It is sufficient to present $\frac{dy}{dx}(4y - \sin x) = y \cos x$ to earn the second point, provided that there are no subsequent errors.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Implicit differentiation
- ☐ Verification

5-7

Solution:

$$\begin{aligned}\frac{d}{dx}(2y^2 - 6) &= \frac{d}{dx}(y \sin x) \Rightarrow 4y \frac{dy}{dx} = \frac{dy}{dx} \sin x + y \cos x \\ \Rightarrow 4y \frac{dy}{dx} - \frac{dy}{dx} \sin x &= y \cos x \Rightarrow \frac{dy}{dx}(4y - \sin x) = y \cos x \\ \Rightarrow \frac{dy}{dx} &= \frac{y \cos x}{4y - \sin x}\end{aligned}$$

Part B

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Any correct tangent line equation will earn the point. No supporting work is required. Simplification of the slope value is not required.



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer

Solution:

At the point $(0, \sqrt{3})$, $\frac{dy}{dx} = \frac{\sqrt{3} \cos 0}{4\sqrt{3} - \sin 0} = \frac{1}{4}$.

An equation for the tangent line is $y = \sqrt{3} + \frac{1}{4}x$.

Part C

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point is earned by any of $\frac{dy}{dx} = 0$, $\frac{y \cos x}{4y - \sin x} = 0$, or $\cos x = 0$.

If additional “correct” x -values are considered outside of the given domain, the response must commit to only $x = \frac{\pi}{2}$ to earn the second point. Any presented y -values, correct or incorrect, are not considered for the second point.

Entering with $x = \frac{\pi}{2}$ does not earn the first point, earns the second point, and is eligible for the third point. The third point is earned for finding $y = 2$. The coordinates do not have to be presented as an ordered pair.

The third point is not earned with additional points present unless the response commits to the correct point.

5-7



0	1	2	3
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The student response accurately includes **all three** of the criteria below.

- ☐ Sets $\frac{dy}{dx} = 0$
- ☐ $x = \frac{\pi}{2}$
- ☐ $y = 2$

Solution:

$$\frac{dy}{dx} = \frac{y \cos x}{4y - \sin x} = 0 \Rightarrow y \cos x = 0 \text{ and } 4y - \sin x \neq 0$$

$$y \cos x = 0 \text{ and } y > 0 \Rightarrow x = \frac{\pi}{2}$$

$$\text{When } x = \frac{\pi}{2}, y \sin x = 2y^2 - 6 \Rightarrow y \sin \frac{\pi}{2} = 2y^2 - 6$$

$$\Rightarrow y = 2y^2 - 6 \Rightarrow 2y^2 - y - 6 = 0$$

$$\Rightarrow (2y + 3)(y - 2) = 0 \Rightarrow y = 2$$

When $x = \frac{\pi}{2}$ and $y = 2$, $4y - \sin x = 8 - 1 \neq 0$. Therefore, the line tangent to the curve is horizontal at the point $(\frac{\pi}{2}, 2)$.

Part D

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point is earned for an attempt to use the quotient rule (or product rule) to find $\frac{d^2y}{dx^2}$.

The second point is earned for correctly finding $\frac{d^2y}{dx^2}$ and evaluating to find that $\frac{d^2y}{dx^2} < 0$ at $(\frac{\pi}{2}, 2)$. The explicit value of $-\frac{2}{7}$ or the equivalent does not need to be reported, but any reported values must be correct in order to earn this point.

The third point can be earned without the second point by reaching a consistent conclusion based on the reported sign of a nonzero value of $\frac{d^2y}{dx^2}$ obtained utilizing $\frac{dy}{dx} = 0$.

Imports: A response is eligible to earn all 3 points in part (d) with a point of the form $(\frac{\pi}{2}, k)$ with $k > 0$, imported from part (c).

5-7



0	1	2	3
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The student response accurately includes **all three** of the criteria below.

- ☐ Considers $\frac{d^2y}{dx^2}$
- ☐ $\frac{d^2y}{dx^2}$ at $(\frac{\pi}{2}, 2)$
- ☐ Answer with justification

Solution:

$$\frac{d^2y}{dx^2} = \frac{(4y - \sin x) \left(\frac{dy}{dx} \cos x - y \sin x \right) - (y \cos x) \left(4 \frac{dy}{dx} - \cos x \right)}{(4y - \sin x)^2}$$

When $x = \frac{\pi}{2}$ and $y = 2$,

$$\begin{aligned} \frac{d^2y}{dx^2} &= \frac{(4 \cdot 2 - \sin \frac{\pi}{2}) (0 \cdot \cos \frac{\pi}{2} - 2 \cdot \sin \frac{\pi}{2}) - (2 \cos \frac{\pi}{2}) (4 \cdot 0 - \cos \frac{\pi}{2})}{(4 \cdot 2 - \sin \frac{\pi}{2})^2} \\ &= \frac{(7)(-2) - (0)(0)}{(7)^2} = \frac{-2}{7} < 0. \end{aligned}$$

f has a relative maximum at the point $(\frac{\pi}{2}, 2)$ because $\frac{dy}{dx} = 0$ and $\frac{d^2y}{dx^2} < 0$.

5-7

18. Let f be a function such that $f(-1) = 1$. At each point (x, y) on the graph of f , the slope is given by $\frac{dy}{dx} = -x^2 - xy + y^2 - 1$. Which of the following statements is true?

- (A) f has a relative minimum at $x = -1$. ✓
- (B) f has a relative maximum at $x = -1$.
- (C) f has neither a relative minimum nor a relative maximum at $x = -1$.
- (D) There is insufficient information to determine whether f has a relative minimum, a relative maximum, or neither at $x = -1$.

Answer A

Correct. The point $(-1, 1)$ is a critical point of f because $\frac{dy}{dx}\big|_{(-1,1)} = -(-1)^2 - (-1)(1) + 1^2 - 1 = 0$. The Second Derivative Test can be used to determine the behavior of the critical point. Using implicit differentiation to calculate the second derivative gives $\frac{d^2y}{dx^2} = \frac{d}{dx}\left(\frac{dy}{dx}\right) = -2x - y - x\frac{dy}{dx} + 2y\frac{dy}{dx}$. Therefore, $\frac{d^2y}{dx^2}\big|_{(-1,1)} = -2(-1) - 1 - (-1)(0) + 2(1)(0) = 1$. Since $\frac{d^2y}{dx^2}\big|_{(-1,1)} > 0$, $x = -1$ is the location of a relative minimum by the Second Derivative Test.

19.  The second derivative of the function f is given by $f''(x) = \sin\left(\frac{x^2}{8}\right) - 2\cos x$. The function f has many critical points, two of which are at $x = 0$ and $x = 6.949$. Which of the following statements is true?

- (A) f has a local minimum at $x = 0$ and at $x = 6.949$.
- (B) f has a local minimum at $x = 0$ and a local maximum at $x = 6.949$.
- (C) f has a local maximum at $x = 0$ and a local minimum at $x = 6.949$.
- (D) f has a local maximum at $x = 0$ and at $x = 6.949$. ✓

Answer D

Correct. The Second Derivative Test may determine whether a critical point is the location of a local maximum or local minimum. Use a calculator to evaluate the second derivative at the two critical points, as follows.

$f''(0) = -2 < 0$ and so $x = 0$ is the location of a local maximum of f .

$f''(6.949) = -1.817 < 0$ and so $x = 6.949$ is the location of a local maximum of f .

5-7

20. The Second Derivative Test cannot be used to conclude that $x = 2$ is the location of a relative minimum or relative maximum for which of the following functions?

(A) $f(x) = \cos(x - 2)$, where $f'(x) = -\sin(x - 2)$

(B) $f(x) = xe^{-\frac{x}{2}}$, where $f'(x) = e^{-\frac{x}{2}} - \frac{1}{2}xe^{-\frac{x}{2}}$

(C) $f(x) = x^2 - 4x - 2$, where $f'(x) = 2x - 4$

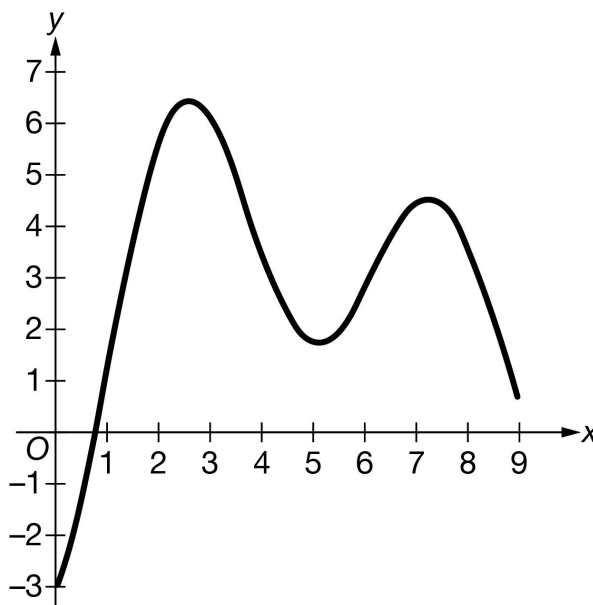
(D) $f(x) = x^3 - 6x^2 + 12x + 1$, where $f'(x) = 3x^2 - 12x + 12$ ✓

Answer D

Correct. For this function, $f''(x) = 6x - 12$. Because $f'(2) = 3(4) - 12(2) + 12 = 0$, $x = 2$ is a critical point. However, $f''(2) = 12 - 12 = 0$, and therefore, the Second Derivative Test is inconclusive about the behavior of the function at the critical point $x = 2$.

5-7

21.

Graph of f''

The graph of f'' , the second derivative of the continuous function f , is shown above on the interval $[0, 9]$. On this interval f has only one critical point, which occurs at $x = 6$. Which of the following statements is true about the function f on the interval $[0, 9]$?

- (A) f has a relative minimum at $x = 6$ but not an absolute minimum.
- (B) The absolute minimum of f is at $x = 6$. ✓
- (C) f has a relative maximum at $x = 6$ but not an absolute maximum.
- (D) The absolute maximum of f is at $x = 6$.

Answer B

Correct. When a continuous function has only one critical point on an interval in its domain and the critical point corresponds to a relative extremum of the function on the interval, then that critical point also corresponds to the absolute extremum of the function on the interval. Since $f''(6) > 0$, the critical point at $x = 6$ is a relative minimum by the Second Derivative Test. Because it is the only critical point on the interval $[0, 9]$, it therefore corresponds to the absolute minimum on the interval.

5-7

22.

x	0	2	4	6	8	10
$f'(x)$	-1	0	-2	3	0	-1
$f''(x)$	8.333	-1.900	0.971	-0.304	0.400	-4.167

Let f be a twice-differentiable function. Selected values of f' and f'' are shown in the table above. Which of the following statements are true?

- I. f has neither a relative minimum nor a relative maximum at $x = 2$.
- II. f has a relative maximum $x = 2$.
- III. f has a relative maximum $x = 8$.

- (A) I only
- (B) II only
- (C) III only
- (D) I and III only

**Answer B**

Correct. Since $f'(2) = 0$ and $f''(2) < 0$, there is a relative maximum at $x = 2$ by the Second Derivative Test. Therefore, statement II is true.

Statement I is false since there is a relative maximum at $x = 2$.

Statement III is false since $f'(8) = 0$ and $f''(8) > 0$ implies that there is a relative minimum at $x = 8$ by the Second Derivative Test, not a relative maximum.

5-7

23. Let f be a twice-differentiable function. Which of the following statements are individually sufficient to conclude that $x = 4$ is the location of the absolute maximum of f on the interval $[0, 10]$?

- I. $f'(4) = 0$
- II. $x = 4$ is the only critical point of f on the interval $[0, 10]$, and $f''(4) < 0$.
- III. $x = 4$ is the only critical point of f on the interval $[0, 10]$, and $f(10) < f(0) < f(4)$.

- (A) II only
- (B) III only
- (C) I and II only
- (D) II and III only



Answer D

Correct. When a continuous function has only one critical point on an interval in its domain and the critical point corresponds to a relative extremum of the function on the interval, then that critical point also corresponds to the absolute extremum of the function on the interval. Since statement II says that $x = 4$ is a critical point of f and $f''(4)$ is negative, it follows from the Second Derivative Test that $x = 4$ is the location of a relative maximum of f . Since statement II also says $x = 4$ is the only critical point of f on the interval $[0, 10]$, this statement is individually sufficient to conclude that $x = 4$ is the location of the absolute maximum of f on the interval $[0, 10]$.

The absolute maximum of a function on a closed interval must occur at a critical point or at an endpoint. Since statement III says that $x = 4$ is the only critical point of f on the interval $[0, 10]$, the only candidates are at $x = 0$, $x = 4$, and $x = 10$. Since statement III also says that $f(10) < f(0) < f(4)$, this statement is individually sufficient to conclude that $x = 4$ is the location of the absolute maximum of f on the interval $[0, 10]$.

Statement I says that $x = 4$ is a critical point of f , but that is not enough to conclude that the critical point is the location of the absolute maximum of f on the interval $[0, 10]$.

5-7

24. Let f be a function such that $f(1) = 2$. At each point (x, y) on the graph of f , the slope is given by $\frac{dy}{dx} = 5xy - x^2 - y^2 - 5$. Which of the following statements is true?

- (A) f has a relative minimum at $x = 1$. ✓
- (B) f has a relative maximum at $x = 1$.
- (C) f has neither a relative minimum nor a relative maximum at $x = 1$.
- (D) There is insufficient information to determine whether f has a relative minimum, a relative maximum, or neither at $x = 1$.

Answer A

Correct. The point $(1, 2)$ is a critical point of f because

$\frac{dy}{dx} \Big|_{(1,2)} = 5(1)(2) - (1)^2 - (2)^2 - 5 = 10 - 1 - 4 - 5 = 0$. The Second Derivative Test can be used to

determine the behavior of the critical point. Using implicit differentiation to calculate the second derivative gives

$\frac{d^2y}{dx^2} = \frac{d}{dx} \left(\frac{dy}{dx} \right) = 5y + 5x \frac{dy}{dx} - 2x - 2y \frac{dy}{dx}$. Therefore,

$\frac{d^2y}{dx^2} \Big|_{(1,2)} = \left(5y + 5x \frac{dy}{dx} - 2x - 2y \frac{dy}{dx} \right) \Big|_{(1,2)} = 10 + 5(0) - 2 - 4(0) = 8$. Since this is positive, $x = 1$ is

the location of a relative minimum by the Second Derivative Test.

25.  The second derivative of the function f is given by $f''(x) = e^{\sin x} (2 \cos x - x \sin x + x \cos^2 x)$. The function f has many critical points, two of which are at $x = 2.074$ and $x = 7.980$. Which of the following statements is true?

- (A) f has local minima at $x = 2.074$ and at $x = 7.980$.
- (B) f has a local minimum at $x = 2.074$ and a local maximum at $x = 7.980$.
- (C) f has a local maximum at $x = 2.074$ and a local minimum at $x = 7.980$.
- (D) f has local maxima at $x = 2.074$ and at $x = 7.980$. ✓

Answer D

Correct. The Second Derivative Test may determine whether a critical point is the location of a local maximum or local minimum. Use a calculator to evaluate the second derivative at the two critical points, as follows.

$f''(2.074) < 0$ so $x = 2.074$ is the location of a local maximum of f .

$f''(7.980) < 0$ so $x = 7.980$ is the location of a local maximum of f .

This tells us that the graph of f is concave down on an interval containing $x = 2.074$ and on an interval containing $x = 7.980$.

5-7

26. The Second Derivative Test cannot be used to conclude that $x = 1$ is the location of a relative minimum or relative maximum for which of the following functions?

(A) $f(x) = \cos(x^2 - 1)$, where $f'(x) = -2x \sin(x^2 - 1)$ and $f''(x) = -2 \sin(x^2 - 1) - 4x^2 \cos(x^2 - 1)$

(B) $f(x) = e^{(x-1)^2}$, where $f'(x) = 2(x-1)e^{(x-1)^2}$ and $f''(x) = 4(x-1)^2 e^{(x-1)^2} + 2e^{(x-1)^2}$

(C) $f(x) = \frac{x^3}{3} + x^2 - 3x + 1$, where $f'(x) = x^2 + 2x - 3$ and $f''(x) = 2x + 2$

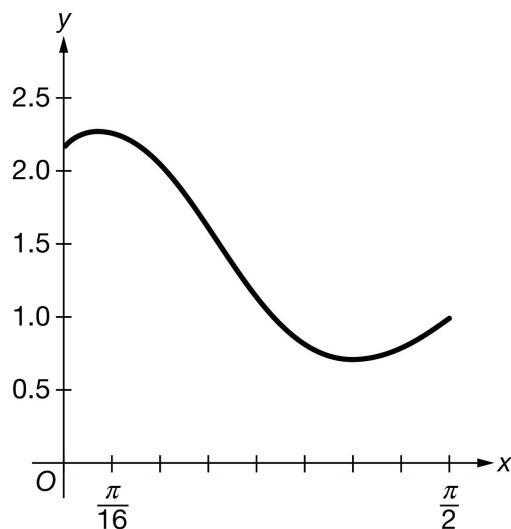
(D) $f(x) = x^4 - 4x^3 + 6x^2 - 4x + 1$, where $f'(x) = 4x^3 - 12x^2 + 12x - 4$ and $f''(x) = 12x^2 - 24x + 12$ ✓

Answer D

Correct. Because $f'(1) = 4 - 12 + 12 - 4 = 0$, $x = 1$ is a critical point. However, $f''(1) = 12 - 24 + 12 = 0$, and therefore, the Second Derivative Test is inconclusive about the behavior of the function at the critical point $x = 1$.

5-7

27.

Graph of f''

The graph of f'' , the second derivative of the continuous function f , is shown above on the interval $[0, \frac{\pi}{2}]$. On this interval f has only one critical point, which occurs at $x = \frac{\pi}{16}$. Which of the following statements is true about the function f on the interval $[0, \frac{\pi}{2}]$?

- (A) f has a relative minimum at $x = \frac{\pi}{16}$ but not an absolute minimum.
- (B) The absolute minimum of f is at $x = \frac{\pi}{16}$. ✓
- (C) f has a relative maximum at $x = \frac{\pi}{16}$ but not an absolute maximum.
- (D) The absolute maximum of f is at $x = \frac{\pi}{16}$.

Answer B

Correct. When a continuous function has only one critical point on an interval in its domain and the critical point corresponds to a relative extremum of the function on the interval, then that critical point also corresponds to the absolute extremum of the function on the interval. Since $f''(\frac{\pi}{16}) > 0$, the critical point at $x = \frac{\pi}{16}$ is a relative minimum by the Second Derivative Test. Because it is the only critical point on the interval $[0, \frac{\pi}{2}]$, it therefore corresponds to the absolute minimum on the interval.

5-7

28.

x	0	1	2	3	4	5
$f'(x)$	-3	0	-1	5	0	-3
$f''(x)$	5.3	-2.0	1.7	-0.5	1.2	-5.1

Let f be a twice-differentiable function. Selected values of f' and f'' are shown in the table above. Which of the following statements are true?

- I. f has neither a relative minimum nor a relative maximum at $x = 1$.
- II. f has a relative maximum at $x = 1$.
- III. f has a relative maximum at $x = 4$.

- (A) I only
- (B) II only
- (C) III only
- (D) I and III only

**Answer B**

Correct. Since $f'(1) = 0$ and $f''(1) < 0$, there is a relative maximum at $x = 1$ by the Second Derivative Test. Therefore, statement II is true.

Statement I is false since there is a relative maximum at $x = 1$.

Statement III is false since $f'(4) = 0$ and $f''(4) > 0$ imply that there is a relative minimum at $x = 4$ by the Second Derivative Test, not a relative maximum.

5-7

29. Let f be a twice-differentiable function. Which of the following statements are individually sufficient to conclude that $x = 2$ is the location of the absolute maximum of f on the interval $[-5, 5]$?

- I. $f'(2) = 0$
- II. $x = 2$ is the only critical point of f on the interval $[-5, 5]$, and $f''(2) < 0$.
- III. $x = 2$ is the only critical point of f on the interval $[-5, 5]$, and $f(-5) < f(5) < f(2)$.

- (A) II only
- (B) III only
- (C) I and II only
- (D) II and III only



Answer D

Correct. When a continuous function has only one critical point on an interval in its domain and the critical point corresponds to a relative extremum of the function on the interval, then that critical point also corresponds to the absolute extremum of the function on the interval. Since statement II says that $x = 2$ is a critical point of f and $f''(2)$ is negative, it follows from the Second Derivative Test that $x = 2$ is the location of a relative maximum of f . Since statement II also says $x = 2$ is the only critical point of f on the interval $[-5, 5]$, this statement is individually sufficient to conclude that $x = 2$ is the location of the absolute maximum of f on the interval $[-5, 5]$.

The absolute maximum of a function on a closed interval must occur at a critical point or at an endpoint. Since statement III says that $x = 2$ is the only critical point of f on the interval $[-5, 5]$, the only candidates are at $x = -5$, $x = 2$, and $x = 5$. Since statement III also says that $f(-5) < f(5) < f(2)$, this statement is individually sufficient to conclude that $x = 2$ is the location of the absolute maximum of f on the interval $[-5, 5]$. Statement I says that $x = 2$ is a critical point of f , but that is not enough to conclude that the critical point is the location of the absolute maximum of f on the interval $[-5, 5]$.

5-7

30. Let g be a twice differentiable function for all x . If $g'(2) = 0$ and $g''(2) < 0$, which of the following statements about g is true?
- (A) The function g has a local minimum at $x = 2$.
- (B) The function g has a local maximum at $x = 2$. ✓
- (C) The graph of g has a point of inflection at $x = 2$.
- (D) The graph of g has a vertical tangent at $x = 2$.

Answer B

Correct. The Second Derivative Test may determine whether a critical point is the location of a local maximum or local minimum. Because $g'(2) = 0$, $x = 2$ is a critical point. Since $g''(2) < 0$, the Second Derivative Test guarantees that g has a local maximum at $x = 2$.

31. If $y = 6x^2 - x^3$, what is the maximum value of the instantaneous rate of change of y with respect to x ?
- (A) 12 ✓
- (B) 16
- (C) 27
- (D) 32

Answer A

Correct. The instantaneous rate of change of y with respect to x is $\frac{dy}{dx} = 12x - 3x^2$. The maximum of $f(x) = 12x - 3x^2$ occurs where $f'(x) = 12 - 6x = 0$, so at $x = 2$. This is the location for the absolute maximum because the graph of f is a parabola opening downward. The maximum value of the instantaneous rate of change is therefore $\left. \frac{dy}{dx} \right|_{x=2} = f(2) = 24 - 12 = 12$.

5-7

32. Let f be the function defined by $f(x) = k\sqrt{x} - \ln x$, for $x > 0$, where k is a positive constant.

(a) Find $f'(x)$ and $f''(x)$.

(b) For what value of the constant k does f have a critical point at $x = 1$? For this value of k , determine whether f has a relative minimum, relative maximum, or neither at $x = 1$. Justify your answer.

(c) For a certain value of the constant k , the graph of f has a point of inflection on the x -axis. Find this value of k .

Part A

1 point is earned for: $f'(x)$

1 point is earned for: $f''(x)$

$$f'(x) = \frac{k}{2\sqrt{x}} - \frac{1}{x}$$

$$f''(x) = -\frac{1}{4}kx^{-3/2} + x^{-2}$$



0	1	2
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The student response earns all of the following points:

1 point is earned for: $f'(x)$

1 point is earned for: $f''(x)$

$$f'(x) = \frac{k}{2\sqrt{x}} - \frac{1}{x}$$

$$f''(x) = -\frac{1}{4}kx^{-3/2} + x^{-2}$$

Part B

1 point is earned if sets $f'(1) = 0$ or $f'(x) = 0$

1 point is earned if solves for k

1 point is earned for the answer

1 point is earned for the justification

5-7

$$f'(1) = \frac{1}{2}k - 1 = 0 \Rightarrow k = 2$$

When $k = 2$, $f'(1) = 0$ and $f''(1) = -\frac{1}{2} + 1 > 0$.

f has a relative minimum value at $x = 1$ by the Second Derivative Test.



0	1	2	3	4
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The student response earns all of the following points:

1 point is earned if sets $f'(1) = 0$ or $f'(x) = 0$

1 point is earned if solves for k

1 point is earned for the answer

1 point is earned for the justification

$$f'(1) = \frac{1}{2}k - 1 = 0 \Rightarrow k = 2$$

When $k = 2$, $f'(1) = 0$ and $f''(1) = -\frac{1}{2} + 1 > 0$.

f has a relative minimum value at $x = 1$ by the Second Derivative Test.

Part C

1 point is earned for: $f''(x) = 0$ or $f(x) = 0$

1 point is earned for the equation in one variable

1 point is earned for the answer

At this inflection point, $f''(x) = 0$ and $f(x) = 0$.

5-7

$$f''(x) = 0 \Rightarrow \frac{-k}{4x^{3/2}} + \frac{1}{x^2} = 0 \Rightarrow k = \frac{4}{\sqrt{x}}$$

$$f(x) = 0 \Rightarrow k\sqrt{x} - \ln x = 0 \Rightarrow k = \frac{\ln x}{\sqrt{x}}$$

$$\text{Therefore, } \frac{4}{\sqrt{x}} = \frac{\ln x}{\sqrt{x}}$$

$$\Rightarrow 4 = \ln x$$

$$\Rightarrow x = e^4$$

$$\Rightarrow k = \frac{4}{e^2}$$



0	1	2	3
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The student response earns all of the following points:

1 point is earned for: $f''(x) = 0$ or $f(x) = 0$

1 point is earned for the equation in one variable

1 point is earned for the answer

At this inflection point, $f''(x) = 0$ and $f(x) = 0$.

$$f''(x) = 0 \Rightarrow \frac{-k}{4x^{3/2}} + \frac{1}{x^2} = 0 \Rightarrow k = \frac{4}{\sqrt{x}}$$

$$f(x) = 0 \Rightarrow k\sqrt{x} - \ln x = 0 \Rightarrow k = \frac{\ln x}{\sqrt{x}}$$

$$\text{Therefore, } \frac{4}{\sqrt{x}} = \frac{\ln x}{\sqrt{x}}$$

$$\Rightarrow 4 = \ln x$$

$$\Rightarrow x = e^4$$

$$\Rightarrow k = \frac{4}{e^2}$$